

Problem Statement

Design a **Patient Health Monitoring and Risk Classification System** using Python and NumPy. The system should analyze multiple patient vital signs and classify health risks based on statistical analysis.

Vital Sign	NumPy Generation	Range
Blood Pressure (BP)	<code>np.random.randint(110, 150, (7, 5))</code>	110 – 149 mmHg
Heart Rate (HR)	<code>np.random.randint(60, 110, (7, 5))</code>	60 – 109 bpm
Oxygen Level (SpO2)	<code>np.random.randint(90, 100, (7, 5))</code>	90 – 99 %

Dataset Details

All data is generated using `np.random.seed(42)` — no manual input required. The three vital signs are simulated as 2D arrays of shape (7 days × 5 patients):

Each **column** = one patient (5 patients total). Each **row** = one day's readings (7 days total).

The system should:

- Store Blood Pressure, Heart Rate, and Oxygen Level data
- Calculate per-patient averages across 7 days (`axis=0`)
- Analyze patient stability using standard deviation across days
- Detect hypertensive patients
- Identify abnormal heart rate and low oxygen levels
- Track multiple risk conditions
- Classify patients into Normal, Moderate Risk, or Critical
- Find the most critical patient using risk score - Use `np.argmax()`

Detect the following risk conditions per patient:

Risk Score	Classification
0 or 1	Normal
2	Moderate Risk
3 or 4	Critical

Compute a risk score (0–4) by summing all triggered conditions

Classify each patient based on risk score:

Condition	Threshold
High BP (Hypertensive)	avg_bp > 130
Low Oxygen	avg_spo2 < 95
Abnormal Heart Rate	avg_hr < 60 or avg_hr > 100
Unstable Vitals	std_bp > 10 or std_hr > 15

Implement the following:

- Use np.random.seed(42) for reproducible data generation
- Use np.mean() and np.std() with axis=0 for per-patient statistics
- Use boolean arrays for each risk condition
- Use .astype(int) to convert boolean flags to integer scores
- Use np.argmax() to find the most critical patient index
- Use a loop with conditional logic for classification labels

Input format :

No user input required. The program uses NumPy random seed 42 to generate fixed data for 7 days x 5 patients.

Output format :

The first line of output prints **Average BP:** followed by the average BP values of 5 patients.

The second line of output prints **Average HR:** followed by the average HR values of 5 patients.

The third line of output prints **Average SpO2:** followed by the average SpO2 values of 5 patients.

The fourth line of output prints **Risk Score:** followed by the risk scores of 5 patients.

The fifth line of output prints **Classification:** followed by the list of classifications.

The sixth line of output prints **Most Critical Patient Index:** followed by the index of the most critical patient.

Refer to the sample output for formatting specifications.

Code constraints :

Fixed random seed = 42

Number of patients = 5

Number of days = 7

Risk score range = 0 to 4 (one point per triggered condition)

Classification has exactly 3 levels: Normal, Moderate Risk, Critical

Sample test cases :

Input 1 :

Output 1 :

Average BP: [131. 133.42857143 133.28571429 126.42857143 126.]

Average HR: [101.85714286 85.71428571 72.42857143 79.57142857 76.71428571]

Average SpO2: [94.71428571 94.85714286 96.42857143 95.85714286 94.28571429]

Risk Score: [4 3 2 0 2]

Classification: ['Critical', 'Critical', 'Moderate Risk', 'Normal', 'Moderate Risk']

Most Critical Patient Index: 0

Note :

The program will be evaluated only after the “Submit Code” is clicked.

Extra spaces and new line characters in the program output will result in the failure of the test case.